

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0772

Peachland 138 kV
November 20, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0772. The Interconnection Customer proposes a 149.4 MW solar photovoltaic facility interconnecting at the Peachland 138 kV substation in the West Virginia area. The request was received on April 08, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 283 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,806,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Drum - Brighton 138 kV L	Artesia Country Club - Air Force Base Substation 345 kV	219.6	211	5.1
Jack McDonough - Low Gap 138 kV	Cark County - Artesia Country Club 345 kV line	189.0	273	11.8
Flint River - Low Gap 138 kV	Low Gap - Canovanas Pueblo 2401 138 kV Line	131.6	182	6.0
Ess Wesel - Pineville Substation 138 kV Line	W Waynesboro Substation - Chambersburg Substation 345 kV L	126.9	277	6.9
Canovanas Pueblo 2401 - Low Gap 138 kV	Low Gap - Flint Riv 138 kV L	111.3	244	8.4
Cressey - Ess Wesel 138 kV	West Waynesboro - Chambersburg Substation 345 kV Line	110.2	351	6.9
Shepherdstown - Lcec No. 5 138 kV L	American - Urban 345 kV line	109.7	338	43.1
Garden Corners - Douglas City 138 kV	Southwest Research Institute - Douglas City 138 kV	104.4	201	9.0
G-P Gypsum - Brighton 138 kV	Rubidoux - Altavista Power Sa 138 kV	102.7	505	5.1
Lamont - University City 138 kV line	Clark County - Kearneyville Substation 345 kV	102.3	291	27.6
Arvin - Garden Corners 138 kV	Douglas City - Southwest Research Institue 138 kV	101.3	171	9.0
Richford - University City 138 kV	Cark County - Kearneyville 345 kV line	100.6	330	27.6
Four Hills - Rubidoux 138 kV L	W Kelso - Dry Gap 138 kV	100.4	540	7.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Columbus City - N Mesa 138 kV Line	Kearneyville - International Paper Valliant Ok 345 kV	100.1	323	17.7
Richford - Sun City 138 kV	Clark County - Kearneyville 345 kV	100.0	345	27.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Brighton - Drum 138 kV	Rebuild line with higher-capacity conductor	\$12,638,000	44% (\$5,538,000)
Low Gap - Jack McDonough 138 kV	Rebuild line with higher-capacity conductor	\$2,222,000	9% (\$193,000)
Low Gap - Flint Riv 138 kV L	Rebuild line with higher-capacity conductor	\$10,821,000	49% (\$5,262,000)
Ess Wesel - Pineville Substation 138 kV Line	Rebuild line with higher-capacity conductor	\$5,175,000	14% (\$716,000)
Canovanas Pueblo 2401 - Low Gap 138 kV	Rebuild line with higher-capacity conductor	\$8,858,000	20% (\$1,765,000)
Cressey - Ess Wesel 138 kV	Rebuild line with higher-capacity conductor	\$5,109,000	14% (\$707,000)
Shepherdstown - Lcec No. 5 138 kV L	Reconductor line	\$1,266,000	57% (\$728,000)
Garden Corners - Douglas City 138 kV	Reconductor line	\$1,004,000	65% (\$655,000)
G-P Gypsum - Brighton 138 kV	Reconductor line	\$2,126,000	42% (\$904,000)
Lamont - University City 138 kV	Reconductor line	\$5,068,000	74% (\$3,761,000)
Arvin - Garden Corners 138 kV	Reconductor line	\$3,566,000	65% (\$2,327,000)
Richford - University City 138 kV	Reconductor line	\$2,021,000	74% (\$1,500,000)
Four Hills - Rubidoux 138 kV L	Reconductor line	\$3,401,000	30% (\$1,010,000)
Columbus City - North Mesa Substation 138 kV Line	Reconductor line	\$517,000	83% (\$430,000)
Richford - Sun City 138 kV	Reconductor line	\$3,112,000	74% (\$2,310,000)
	Total allocated to project		\$27,806,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.